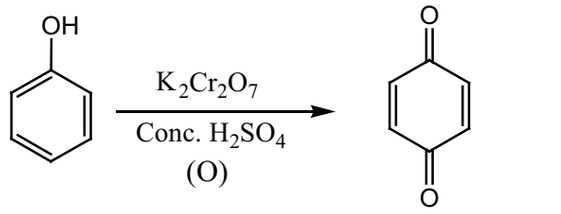


Phenol Sodium Phenoxide Anisole (methoxybenzene)

d) Oxidation:

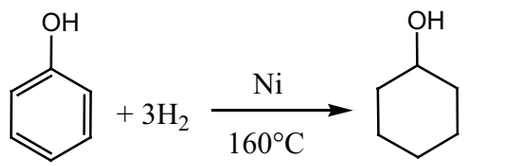
Phenol undergoes oxidation with air or acidified $K_2Cr_2O_7$ with conc. H_2SO_4 to form 1,4-benzoquinone.



Phenol 1,4-benzoquinone

e) Reduction:

Phenol on catalytic hydrogenation gives cyclohexanol.



Phenol cyclohexanol

Reactions of benzene ring:

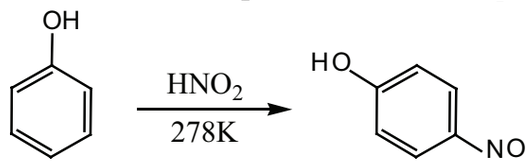
Electrophilic aromatic substitution:

We have already learnt in XI standard that the groups like -$\ddot{O}H$, -$\ddot{N}H_2$, etc., which when directly attached to the benzene ring, activate the ring towards electrophilic substitution reaction and direct the incoming electrophile to occupy either the ortho or para position.

Common electrophilic aromatic substitutions are as follows:

i) Nitrosation:

Phenol can be readily nitrosoated at low temperature with HNO_2.



phenol p-nitroso phenol

XII U11-Hydroxy compounds.indd 128 16-12-2022 19:42:13